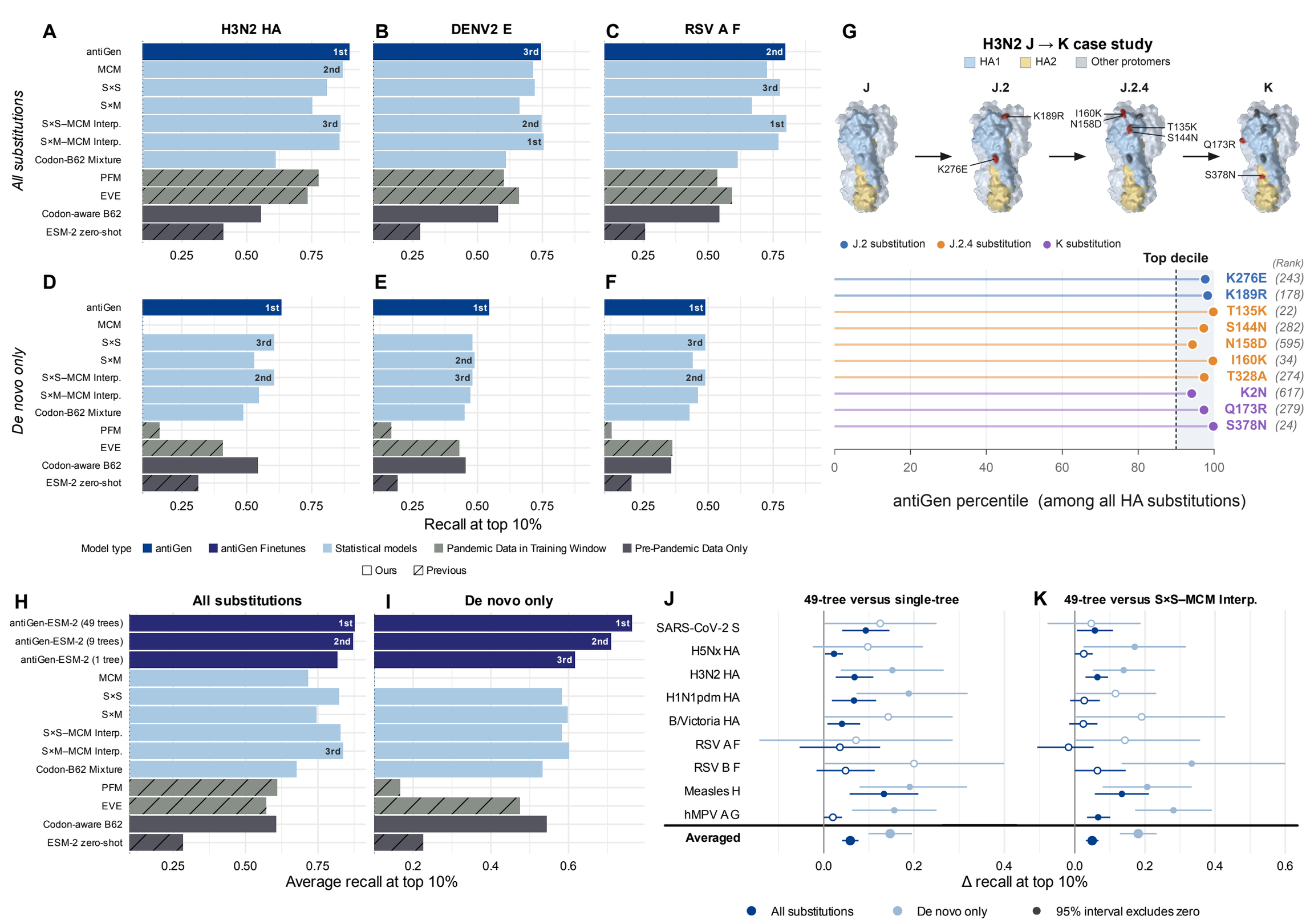

A
H3N2 HA
B
DENV2 E
C
RSV A F
All substitutions
De novo only
antiGen
MCM
S×S
S×M
S×S–MCM Interp.
S×M–MCM Interp.
Codon-B62 Mixture
PFM
EVE
Codon-aware B62
ESM-2 zero-shot
1st
2nd
3rd
0.25
0.50
0.75
D
E
F
Recall at top 10%
Model type
antiGen
antiGen Finetunes
Statistical models
Pandemic Data in Training Window
Pre-Pandemic Data Only
Ours
Previous
G
H3N2 J → K case study
HA1
HA2
Other protomers
J
J.2
J.2.4
K
K189R
K276E
I160K
N158D
T135K
S144N
Q173R
S378N
J.2 substitution
J.2.4 substitution
K substitution
Top decile
(Rank)
K276E (243)
K189R (178)
T135K (22)
S144N (282)
N158D (595)
I160K (34)
T328A (274)
K2N (617)
Q173R (279)
S378N (24)
0
20
40
60
80
100
antiGen percentile (among all HA substitutions)
H
All substitutions
antiGen-ESM-2 (49 trees)
antiGen-ESM-2 (9 trees)
antiGen-ESM-2 (1 tree)
I
De novo only
Average recall at top 10%
J
49-tree versus single-tree
SARS-CoV-2 S
H5Nx HA
H3N2 HA
H1N1pdm HA
B/Victoria HA
RSV A F
RSV B F
Measles H
hMPV A G
Averaged
Δ recall at top 10%
K
49-tree versus S×S–MCM Interp.
All substitutions
De novo only
95% interval excludes zero